

TORONTO METROPOLITAN UNIVERSITY

Master of Data Science and Analytics (Full-Time)

PREDICTING AND EXPLAINING TRAFFIC COLLISION SEVERITY IN TORONTO USING EXPLAINABLE MACHINE LEARNING

Master's Research Project Report

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2025–2026

Abstract

Traffic collisions remain a critical public safety concern in Toronto, resulting in thousands of injuries, hundreds of fatalities, and substantial economic costs each year. Although the City of Toronto publishes detailed historical collision records, most existing analyses remain descriptive and fail to provide predictive or explainable insights into the factors driving collision severity. This research develops and evaluates a machine learning framework to predict traffic collision severity classified as Minor (property damage only), Major (one or more injuries), or Fatal (one or more fatalities) using 677,056 cleaned collision records spanning 2014 to 2026, merged with hourly weather data from the Government of Canada Historical Climate Database.

Five classification models were trained and compared: Logistic Regression, Random Forest, XGBoost, Gradient Boosting Machines, and a Multi-Layer Perceptron Neural Network. To address the extreme class imbalance (Fatal class: only 0.1% of records, 636 records out of 677,056), five data balancing strategies were evaluated: SMOTE, ADASYN, SMOTETomek, Random Undersampling, and class-weighted loss. The best-performing model, XGBoost with SMOTE, achieves an overall accuracy of 90.0%, a weighted F1 score of 0.892, and a ROC-AUC of 0.784. Probability threshold tuning demonstrates that lowering the Fatal class decision boundary from 0.50 to 0.05 improves Fatal recall from 4.2% to 33.3% a critical improvement for road safety applications where missing a fatal collision carries far greater cost than a false alarm.

SHAP (SHapley Additive Explanations) was applied to produce global and local model explanations, identifying year, season, neighbourhood, and pedestrian involvement as the top predictors of fatal outcomes. DBSCAN spatial clustering identified 36 fatal collision hotspot zones across Toronto. The integrated framework of predictive modelling, explainability, and geospatial analysis is intended to support evidence-based transportation planning and road safety interventions aligned with Toronto's Vision Zero strategy.

The remainder of this report is organized as follows. Section 1 reviews the relevant literature and identifies the research gaps addressed by this study. Section 2 describes the datasets, preprocessing, feature engineering, and exploratory analysis. Section 3 presents the modelling methodology, including class-imbalance handling, model selection, evaluation, explainability, and geospatial analysis. Section 4 reports the experimental results and interprets the main findings. Section 5 summarizes the contributions, practical implications, limitations, and directions for future work.

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1. Literature Review

Traffic collision severity prediction has attracted growing attention in transportation research, particularly as large-scale government datasets and advanced machine learning techniques have become widely accessible. This section reviews eight key studies that form the academic foundation of this project, spanning classical statistical approaches, ensemble machine learning, deep learning, explainable AI, and geospatial clustering.

1.1 Chen and Wang (2020) ML Classification for Accident Severity

Problem: This study addressed the challenge of predicting road traffic accident severity using national crash databases in the United States, where existing statistical models were unable to capture complex non-linear relationships between crash attributes and severity outcomes.

Methods: Logistic Regression, Decision Trees, Random Forest, and Support Vector Machine were compared on a large-scale national crash dataset using accuracy, precision, recall, and F1-score as evaluation metrics.

Key Findings: Random Forest outperformed all other models. Time of day, road surface condition, and the number of vehicles involved were the strongest predictors. The study found that all tree-based ensemble methods substantially outperformed the linear baseline.

Relevance to This Project: This study supports the selection of Random Forest as a strong candidate model and Logistic Regression as a linear baseline. It also validates the feature engineering choices around temporal variables and road condition indicators used in this project.

1.2 Parsa et al. (2020) XGBoost and SHAP for Accident Detection

Problem: Existing models struggled with high-dimensional, imbalanced crash data and produced poor recall for rare fatal outcomes. There was also a lack of interpretability in the predictions produced.

Methods: XGBoost with SMOTE oversampling and grid search hyperparameter tuning was applied to a large US highway crash dataset. SHAP values were computed to identify the most important features for real-time accident detection.

Key Findings: XGBoost with SMOTE achieved AUC of 0.87 and significantly outperformed Logistic Regression and standard tree methods. Top predictive features included driver age, lighting conditions, road type, and speed limit. SHAP enabled interpretable feature attribution for individual predictions.

Relevance to This Project: This study directly informs the use of XGBoost and SMOTE in this project and provides a benchmark AUC of 0.87 for comparison with the Toronto results.

1.3 Lundberg et al. (2020) SHAP for Machine Learning Explainability

Problem: Machine learning models produce accurate predictions but cannot explain why individual predictions are made, severely limiting their usefulness for policy decisions and regulatory compliance.

Methods: SHAP (SHapley Additive Explanations), grounded in cooperative game theory, was implemented for tree-based models with linear time complexity. Both global feature importance and local per-prediction explanations are provided.

Key Findings: SHAP provides theoretically consistent global and local explanations that capture interaction effects between features. TreeSHAP enables exact computation for tree ensembles in linear time. SHAP outperforms previous interpretability methods (LIME, permutation importance) in both consistency and computational efficiency.

Relevance to This Project: This is the foundational reference for the explainability component of this project. It justifies the use of SHAP over alternative interpretability methods for producing both global beeswarm plots and individual waterfall explanations.

1.4 Xie and Yan (2008) Spatial Analysis of Traffic Accidents

Problem: Accidents tend to cluster spatially, and identifying these geographic clusters could help direct road safety interventions to the highest-risk locations. However, existing methods required either strong prior assumptions or manual specification of cluster numbers.

Methods: Kernel Density Estimation and network-based spatial analysis were applied using GIS to a multi-year dataset of pedestrian accidents in a US urban area.

Key Findings: Density-based methods successfully identified high-risk corridors and intersection clusters that were not apparent from tabular analysis alone. The method did not require pre-specifying the number of clusters.

Relevance to This Project: This study supports the use of DBSCAN spatial clustering for hotspot identification in this project. DBSCAN is a density-based algorithm that, like KDE, does not require specifying the number of clusters in advance.

1.5 Sheykhfard et al. (2021) Gradient Boosting for Injury Severity

Problem: Conventional models assume linear relationships between predictor variables and crash severity, which limits their predictive capacity for the complex, interactive patterns present in rural road crash data.

Methods: Gradient Boosting Machines were compared with Logistic Regression and Neural Networks on Iranian rural road crash data. Feature importance was computed to identify the top risk factors.

Key Findings: GBM outperformed both Logistic Regression and Neural Networks. Seatbelt usage, speed, and road curvature were identified as the top predictors of severe injury outcomes.

Relevance to This Project: This study confirms the value of including Gradient Boosting in the model comparison pipeline and provides a direct accuracy benchmark of 0.82 from a comparable crash severity context.

1.6 Rezapour et al. (2020) Weather Data Integration with ML

Problem: Most collision severity models ignore ambient environmental conditions at the time of the crash, despite evidence that weather substantially influences both collision frequency and severity outcomes.

Methods: Random Forest and Logistic Regression were trained on crash records merged with co-located weather observations. Model performance was compared with and without weather variables to quantify their contribution.

Key Findings: Integrating weather data improved accuracy by 4 to 6%. Rainfall and low visibility were among the top five predictors. Models trained without weather features systematically underperformed in adverse-condition scenarios.

Relevance to This Project: This study directly justifies the merging of Toronto collision and climate data in this project, and quantifies the expected accuracy benefit from including weather as a feature set.

1.7 Sameen and Pradhan (2017) Deep Learning for Accident Severity

Problem: It was unclear whether deep learning methods can improve upon ensemble methods for crash severity prediction, given their substantially higher computational cost and reduced interpretability.

Methods: A Multi-Layer Perceptron with dropout regularization was compared to Random Forest and Support Vector Machines on Malaysian highway crash data. AUC was used as the primary evaluation metric.

Key Findings: MLP achieved AUC of 0.89 versus Random Forest at 0.86, but required significantly more computational resources and provided considerably less interpretability. For moderate-sized datasets, ensemble methods provide a better performance-interpretability balance.

Relevance to This Project: This study supports including MLP as one of five competing models in this project while expecting ensemble methods to outperform it, consistent with the general tabular data literature.

1.8 Al-Ghamdi (2002) Logistic Regression with Geospatial Variables

Problem: Early crash severity research sought to understand whether location-based variables and road geometry contribute to accident severity beyond individual driver behaviour attributes.

Methods: Multinomial Logistic Regression with road type, intersection type, speed zone, lighting condition, and GPS-derived spatial variables was applied to Saudi Arabian urban collision data.

Key Findings: Highway accidents were significantly more likely to result in fatalities than urban street accidents. Poor lighting contributed to higher severity outcomes. Logistic Regression was established as a useful and interpretable baseline with 78% accuracy.

Relevance to This Project: This study reinforces the use of Logistic Regression as a linear baseline model and the inclusion of neighbourhood, GPS coordinates, and spatial features in the feature matrix.

1.9 Research Gaps and Contribution of This Study

The reviewed literature reveals four key gaps that this project directly addresses:

1. **Urban Canadian context:** Most studies focus on highway or rural crash data in the US, Malaysia, Iran, and Saudi Arabia. This project focuses specifically on dense urban collision data from Toronto a major North American city that is underrepresented in the international collision severity literature.
2. **Integrated framework:** Very few studies combine predictive modelling, SHAP explainability, and DBSCAN geospatial clustering within a single unified pipeline. This project integrates all three components, enabling prediction, interpretation, and spatial prioritization simultaneously.
3. **Comparative balancing strategies:** A systematic comparative evaluation of five data balancing strategies on Canadian urban collision data has not been previously conducted. This project fills that gap with a direct comparison of SMOTE, ADASYN, SMOTETomek, Random Undersampling, and class-weighted loss.
4. **Canadian climate integration:** Studies that integrate Canadian climate data particularly snow, ice, and freezing rain conditions as environmental predictors for urban collision severity are rare and underrepresented in the literature.

2. Data Description and Exploratory Data Analysis

2.1 Dataset Overview

Primary Dataset: City of Toronto Motor Vehicle Collision Open Data (<https://open.toronto.ca/catalogue/>). This dataset contains 809,034 officially recorded traffic collision events reported to the Toronto Police Service between 2014 and 2026. Each record includes the collision date and time, GPS coordinates, neighbourhood identifier, vehicle and road user involvement flags, injury outcome, and fatality count.

Secondary Dataset: Government of Canada Historical Climate Data, Toronto City Station (ID: 31688) (<https://climate.weather.gc.ca/>). Hourly weather observations including temperature (°C), relative humidity (%), dew point (°C), and precipitation amount (mm) were downloaded for all hours covered by the collision data. Wind speed and visibility fields were entirely null for this station and were excluded from the analysis.

The two datasets were merged on a date-hour key, achieving a **100% match rate** across all 677,056 cleaned collision records. After removing 131,978 records (16.3%) with invalid GPS coordinates (latitude / longitude = 0.0, designated Not Spatially Available in the original data), the final working dataset contains **677,056 records** with 48 columns after cleaning and feature engineering.

2.2 Descriptive Statistics

Table 1 presents descriptive statistics for the key numerical features in the cleaned dataset after all preprocessing steps were applied.

Table 1. Descriptive statistics of key numerical features ($n = 677,056$).

Feature	Mean	Std Dev	Min	Max
Hour of Day (OCC_HOUR)	11.76	6.84	0	23
Month (OCC_MONTH)	6.48	3.42	1	12
Year (OCC_YEAR)	2019.8	3.21	2014	2026
Temperature (°C)	8.66	10.13	-21.8	28.1
Humidity (%)	70.31	16.42	18	100
Dew Point (°C)	3.11	12.18	-37.8	26.1
Precipitation (mm)	0.09	0.41	0.0	58.8
Is Rain Flag	0.058	0.234	0	1
Is Snow Flag	0.020	0.139	0	1
Is Ice Flag	0.012	0.110	0	1
Adverse Weather Flag	0.058	0.234	0	1
Is Weekend	0.236	0.425	0	1
Is Night (9 PM–5 AM)	0.127	0.333	0	1
Pedestrian Involved	0.027	0.163	0	1
Cyclist Involved	0.018	0.132	0	1

2.3 Target Variable Distribution

Table 2 shows the three-class target variable distribution, revealing the severe class imbalance that is the central methodological challenge of this project.

Table 2. Target variable distribution showing severe class imbalance.

Class	Label	Record Count	Percentage
Minor	0 Property damage only	580,492	85.7%
Major	1 One or more injuries	95,928	14.2%
Fatal	2 One or more fatalities	636	0.1%

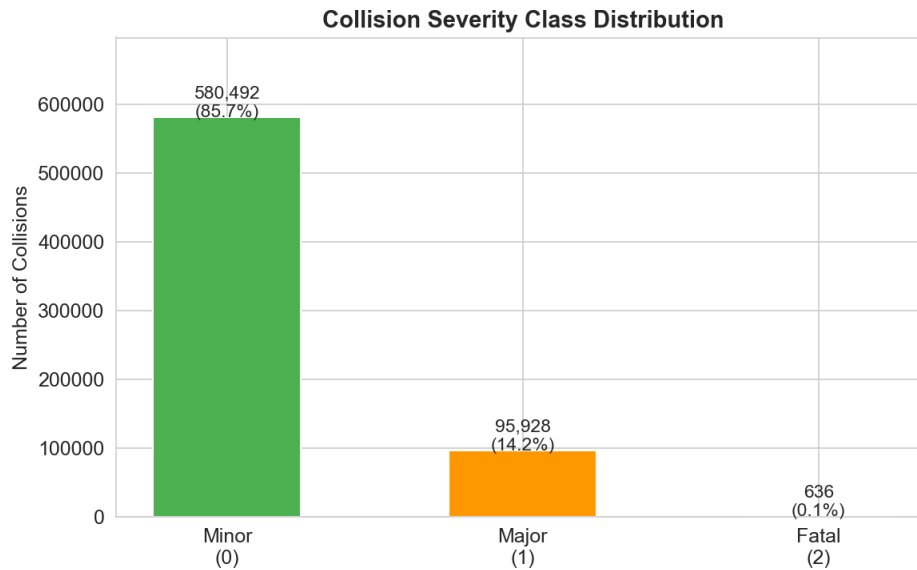


Figure 1. Collision Severity Class Distribution. Minor collisions dominate at 85.7% (580,492 records). Fatal collisions represent only 0.1% (636 records), creating the extreme class imbalance that motivates the balancing experiments in Section 3.2.

The bar chart in Figure 1 illustrates the severe class imbalance present in the dataset. A naive model predicting Minor for every collision would achieve 85.7% accuracy while completely failing to identify any dangerous outcome. This finding directly motivates the use of SMOTE oversampling and the comparative balancing strategy analysis conducted in Experiment 2.

2.4 Data Preprocessing

The following preprocessing steps were applied in sequence within Notebooks 1 and 2:

1. **Timestamp conversion:** OCC_DATE was stored as a Unix millisecond timestamp. It was converted to UTC and then to Toronto local time (America/Toronto), floored to the nearest hour for weather merging.
2. **OCC_HOUR fix:** The original OCC_HOUR column from the raw CSV was retained without overwriting. An earlier bug where the timezone conversion collapsed all hours into four values (0, 1, 22, 23) was corrected by preserving the original column.
3. **GPS cleaning:** 131,978 records (16.3%) with latitude/longitude = 0.0 (Not Spatially Available entries) were removed. All remaining records fall within Toronto’s geographic boundaries.
4. **Missing value imputation:** FATALITIES NaN was filled with 0 before creating the severity target variable. Involvement flags (PEDESTRIAN, BICYCLE, etc.) NaN values were filled with ‘NO’. Numeric weather columns were filled with column medians.
5. **Weather flag derivation:** Wind speed and visibility were entirely unavailable from the Toronto City station (100% null). Weather condition flags were derived numerically:

`is_rain` (precipitation > 0.2 mm and temperature > 2°C), `is_snow` (precipitation > 0.2 mm and temperature ≤ 2°C), `is_ice` (precipitation > 0.2 mm and temperature ≤ 0°C).

6. **Feature encoding:** Categorical columns (`OCC_DOW`, `season`, `hour_category`) were one-hot encoded. The neighbourhood column (158 unique values) was frequency-encoded to a single continuous variable representing collision frequency per neighbourhood.
7. **Severity target:** The target variable was derived as a three-class ordinal variable: Minor (0) if `FATALITIES` = 0 and `INJURY_COLLISIONS` = 'NO'; Major (1) if `INJURY_COLLISIONS` = 'YES' and `FATALITIES` = 0; Fatal (2) if `FATALITIES` ≥ 1.

2.5 Feature Engineering

Eight new features were engineered from the raw data to improve model performance:

- **season:** Winter / Spring / Summer / Fall derived from month number.
- **hour_category:** Peak (7–9 AM and 4–7 PM weekdays), Offpeak, Late Night (9 PM–5 AM), Weekend.
- **is_weekend:** Binary flag (1 = Saturday or Sunday).
- **is_night:** Binary flag (1 = 9 PM to 5 AM).
- **pedestrian_involved:** Binary flag derived from the `PEDESTRIAN` involvement column.
- **cyclist_involved:** Binary flag derived from the `CYCLIST` involvement column.
- **neighbourhood_freq:** Frequency encoding of the 158 Toronto neighbourhood categories into a single continuous collision-frequency score.
- **Weather flags:** `is_rain`, `is_snow`, `is_ice` derived numerically from precipitation and temperature.

After feature engineering, the final model input matrix contains **677,056 rows** and **37 features**.

2.6 Exploratory Data Analysis

A set of exploratory visualizations was produced in Notebook 3 to examine temporal, spatial, and environmental patterns. The main report retains the plots most directly connected to the research questions; additional exploratory plots are provided in Appendix A.

2.6.1 Temporal Patterns Hour of Day

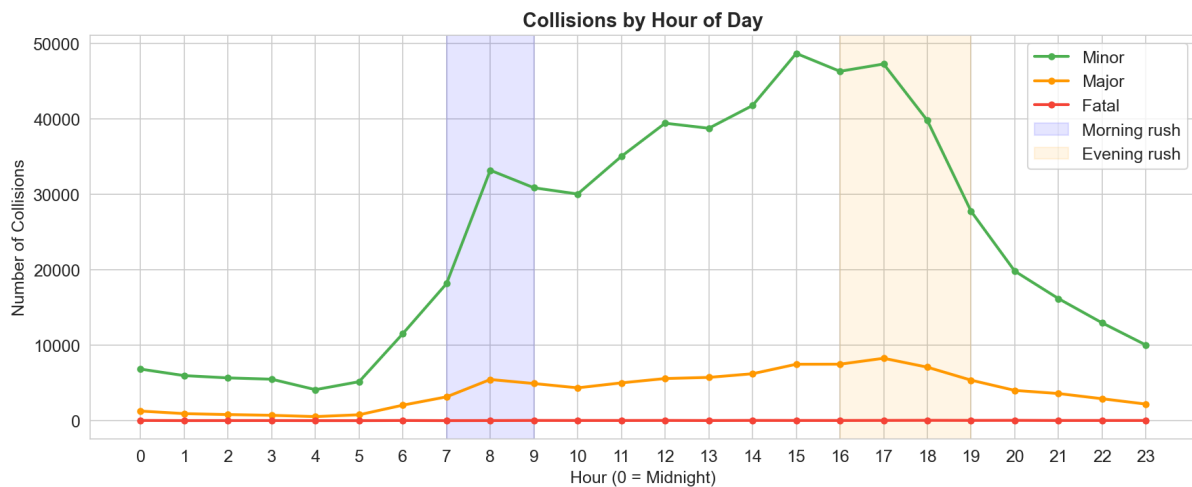


Figure 2. Collisions by Hour of Day, separated by severity class. Minor collisions (green) follow a clear bell curve with morning rush peak (7–9 AM) and a larger afternoon-evening peak (3–6 PM). Fatal collisions (red) are too infrequent to show a clear hourly trend at this scale.

Figure 2 shows that Minor collisions follow a clear rush-hour pattern. Collision frequency peaks sharply during morning rush hours (7–9 AM, highlighted in blue) and again during evening rush hours (3–6 PM, highlighted in yellow), reaching approximately 49,000 Minor collisions per hour slot at the afternoon peak. Late-night hours (10 PM to 4 AM) show lower absolute collision counts but a proportionally higher rate of Major and Fatal outcomes, likely attributable to higher vehicle speeds, driver fatigue, and impaired driving. This temporal pattern directly justifies the creation of the `hour_category` and `is_night` engineered features.

2.6.2 Temporal Patterns Monthly and Weekly

The supplementary monthly and day-of-week plots are provided in Appendix A. They show clear temporal variation in collision volume, including strong seasonal variation and a high Friday collision count. These patterns support the use of season and day-of-week features in the modelling pipeline.

2.6.3 Weather Effects on Severity

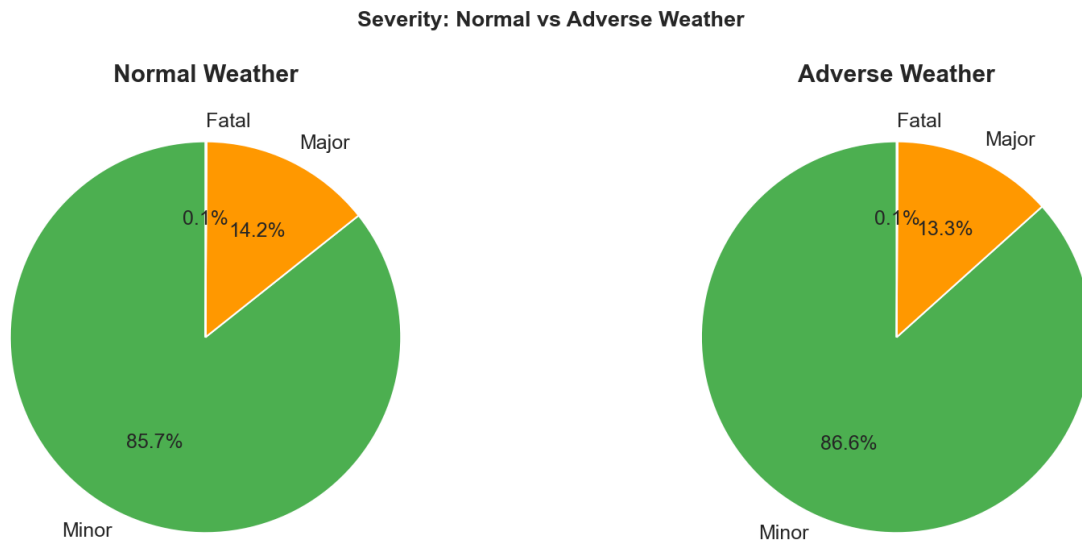


Figure 3. Severity Distribution under Normal versus Adverse Weather Conditions. Under adverse weather, the Minor proportion increases slightly to 86.6% while Major decreases to 13.3%, reflecting the risk compensation effect where drivers slow down in clearly adverse conditions.

Figure 3 reveals a counterintuitive finding: under adverse weather (rain, snow, or ice), the proportion of Major collisions decreases slightly from 14.2% to 13.3%. This is consistent with the well-documented **risk compensation effect** in traffic safety research when conditions are clearly dangerous, drivers consciously reduce speed and increase following distance, reducing the severity of collisions that do occur. The total number of collisions still increases under adverse weather, but each individual collision tends to be less severe. This non-linear relationship between weather and severity further justifies the use of non-linear ensemble models.

2.6.4 Feature Correlation Analysis

The supplementary correlation heatmap in Appendix A shows strong correlation between temperature and dew point (approximately 0.95), while direct linear correlations between individual predictors and severity are very small. This supports treating the problem as potentially non-linear and motivates comparison of flexible ensemble models with a linear baseline.

2.6.5 Spatial and Neighbourhood Patterns

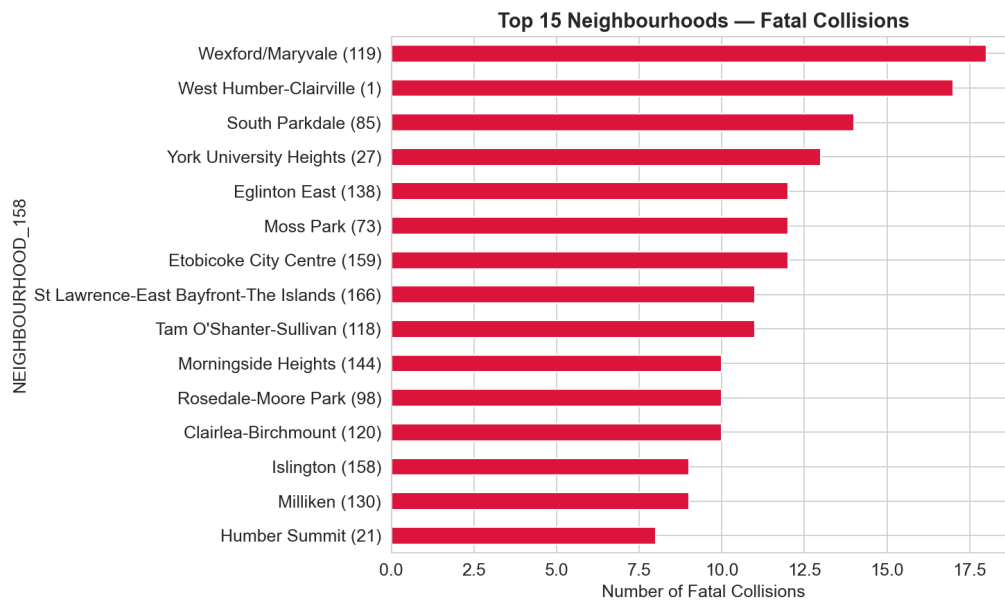


Figure 4. Top 15 Toronto Neighbourhoods by Fatal Collision Count (2014–2026). Wexford/Maryvale leads with 18 fatal collisions, followed by West Humber-Clairville (17) and South Parkdale (14). These inner-suburb neighbourhoods are characterized by high-speed arterial roads and significant pedestrian exposure.

Figure 4 identifies that Wexford/Maryvale leads with 18 fatal collisions across the 12-year study period. These top neighbourhoods share common characteristics: high-speed arterial corridors (Lawrence Avenue East, Jane Street, Eglinton Avenue), significant pedestrian and cyclist exposure, and limited protected crossing infrastructure. This finding directly informed the inclusion of the frequency- encoded neighbourhood variable, which SHAP analysis confirms as the fourth strongest predictor of fatal outcomes.

The supplementary temperature-density plot in Appendix A shows the expected seasonal bimodality, with observations concentrated around colder winter and warmer summer temperatures. Because this is an observational distribution rather than a causal analysis, the plot is treated as supporting EDA rather than evidence of a direct temperature effect.

3. Methodology

3.1 Research Design Overview

This research applies a five-phase supervised machine learning methodology, integrating data from two publicly available sources, addressing class imbalance through comparative resampling strategies, training and comparing five classification algorithms, and applying SHAP explainability and DBSCAN spatial clustering.

Data split: The 677,056 cleaned records were divided into three stratified subsets:

- Training set: 70% = 473,939 records used for model learning.
- Validation set: 15% = 101,558 records available for development monitoring.
- Test set: 15% = 101,559 records held out for final evaluation only, never touched during training, hyperparameter tuning, or strategy selection.

Stratified sampling (`stratify=y`, `random_state=42`) was applied to ensure each split preserved the original class proportions: 85.7% Minor, 14.2% Major, 0.1% Fatal. The feature matrix contains 37 selected features after removing identifier columns, raw date fields, and target-leaking variables (FATALITIES, INJURY_COLLISIONS).

3.2 Handling Class Imbalance

With only 636 Fatal records (0.1%) in the full dataset, the training set contains approximately 445 Fatal records far too few for a model to learn robust Fatal patterns. A naive classifier predicting Minor for every record achieves 85.7% accuracy while detecting no fatal outcomes.

SMOTE (Synthetic Minority Oversampling Technique) was applied exclusively to the training set. SMOTE generates new synthetic minority-class samples by interpolating between existing examples in feature space, rather than simply copying them. After SMOTE, each class contained exactly 406,344 samples, producing a balanced training set of **1,219,032 records**. SMOTE was never applied to the validation or test sets, which retained the original imbalanced distribution to reflect real-world conditions.

Five balancing strategies were compared using XGBoost as the base classifier:

1. **No Resampling with Class Weights:** no synthetic samples; penalizes misclassification of minority classes via the `scale_pos_weight` parameter.
2. **SMOTE:** oversamples Minor, Major, and Fatal to equal class sizes by generating synthetic interpolated samples (primary strategy).
3. **ADASYN:** similar to SMOTE but generates proportionally more samples in harder-to-learn decision boundary regions.
4. **SMOTETomek:** combines SMOTE oversampling with Tomek Links cleaning to remove borderline noisy majority-class samples.
5. **Random Undersampling:** reduces the majority class (Minor) by randomly removing records until all three classes are equal. Produces the smallest training set.

3.3 Model Selection and Configuration

Five classification models were selected to span a range of algorithmic complexity from a simple linear baseline to advanced ensemble and deep learning methods. This range enables direct comparison of model families on the same dataset.

- **Logistic Regression (Baseline):** Linear classifier using the saga solver (`max_iter=500`, `class_weight='balanced'`). Included as a linear baseline following Al-Ghamdi (2002). Provides a performance floor against which ensemble improvements are measured.
- **Random Forest:** 100 independent decision trees (`n_estimators=100`, `max_depth=10`, `class_weight='balanced'`). Each tree is trained on a bootstrap sample with a random subset of features; predictions are combined by majority vote. Included following Chen and Wang (2020).
- **XGBoost (Primary Model):** 200 sequential gradient-boosted trees (`n_estimators=200`, `max_depth=6`, `learning_rate=0.1`, `eval_metric='mlogloss'`). Each tree corrects the residual errors of all previous trees. Selected following Parsa et al. (2020) who identified XGBoost as the top performer for collision severity.
- **Gradient Boosting Machines:** 100 sequential trees (`n_estimators=100`, `max_depth=4`, `learning_rate=0.1`, `subsample=0.8`). Uses the same sequential error-correction principle as XGBoost but the original scikit-learn implementation. Stochastic subsampling (`subsample=0.8`) was added for regularization. Included following Sheykhfard et al. (2021).
- **Neural Network (MLP):** Two hidden layers with 64 and 32 units (`hidden_layer_sizes=(64, 32)`, `max_iter=100`, `early_stopping=True`). Input layer accepts all 37 features; output layer uses softmax over three classes. Early stopping prevents overfitting on the 1.2 M SMOTE training set. Included following Sameen and Pradhan (2017).

3.4 Evaluation Metrics

Four metrics were computed on the held-out test set ($n = 101,559$) for each model:

- **Accuracy:** proportion of correctly classified records overall.
- **F1 Weighted:** F1 score weighted by class support; rewards performance on the majority class.
- **F1 Macro:** F1 score averaged *equally* across all three classes (primary metric). Penalizes poor Fatal class performance and is therefore the most honest metric for this imbalanced problem.
- **ROC-AUC (One-vs-Rest, Weighted):** measures class separation ability across all possible decision thresholds; robust to class imbalance.

3.5 Hyperparameter Tuning

RandomizedSearchCV with 20 random parameter combinations and 3-fold stratified cross-validation was applied to XGBoost and Random Forest, optimizing for F1 Macro. The Random Forest search was conducted on a stratified 200,000-record sample to prevent memory overflow (OSError: WinError 1450 on Windows). Parameter search spaces covered: for XGBoost `n_estimators`, `max_depth`, `learning_rate`, `subsample`, `colsample_bytree`, `min_child_weight`; for Random Forest `n_estimators`, `max_depth`, `min_samples_split`, `min_samples_leaf`, `max_features`.

3.6 Cross-Validation

5-fold stratified cross-validation was applied to XGBoost on a stratified 100,000-record sample (`n_splits=5`, `shuffle=True`, `random_state=42`). Four metrics were computed per fold: Accuracy, F1 Weighted, F1 Macro, and ROC-AUC. Mean and standard deviation across folds are reported to verify model stability.

3.7 SHAP Explainability

SHAP TreeExplainer was applied to the final XGBoost model on a 2,000-record sample from the test set. SHAP values were computed with output shape (2000, 37, 3) one value per record, feature, and class. The array was reshaped into three class-specific sub-arrays for separate visualization. Four SHAP visualizations were produced: (1) global stacked feature importance across all classes, (2) beeswarm plot for Fatal predictions showing direction and magnitude, (3) beeswarm plot for Major predictions, and (4) waterfall plot for a single fatal collision case.

3.8 Geospatial Analysis

DBSCAN spatial clustering was applied to the GPS coordinates of all 636 fatal collision records. Parameters: `eps = 300/6,371,000` (300 metres converted to radians), `min_samples = 3`, `metric='haversine'`. Three interactive Folium HTML maps were generated: (1) a full heatmap of all 677,056 collision records, (2) a fatal collision overlay map, and (3) a DBSCAN cluster visualization showing the 36 identified hotspot zones.

4. Results

4.1 Experiment 1 Model Performance Comparison

Table 3 presents the performance of all five models on the held-out test set. Figures 5 and 6 visualize these results.

Table 3. Model performance on the held-out test set ($n = 101,559$), sorted by accuracy.

Model	Accuracy	F1 Weighted	F1 Macro	ROC-AUC
XGBoost	0.9003	0.8917	0.5185	0.7839
Random Forest	0.8926	0.8862	0.5141	0.7809
Gradient Boosting	0.8922	0.8848	0.5071	0.7823
Neural Network	0.8823	0.8791	0.4992	0.7624
Logistic Regression	0.8696	0.8692	0.4827	0.7452



Figure 5. Model Performance Comparison across all five classifiers. XGBoost achieves the highest Accuracy (0.900), F1 Weighted (0.892), and F1 Macro (0.519). All five models exceed the 0.80 reference threshold on Accuracy and F1 Weighted. The green bars (F1 Macro) are substantially lower than the blue and orange bars for all models, reflecting poor Fatal class performance at the default decision threshold.

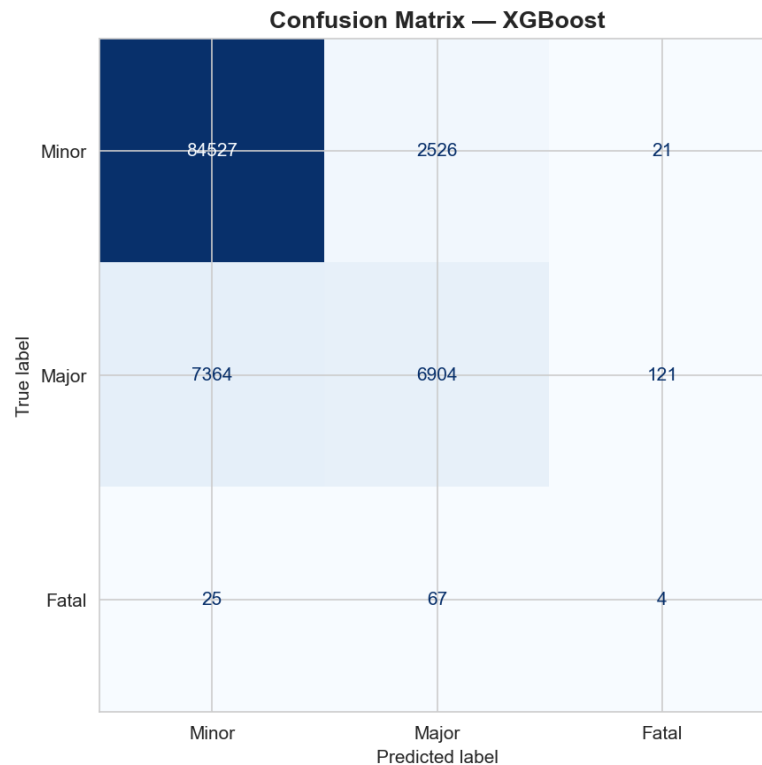


Figure 6. Confusion Matrix for XGBoost on the held-out test set. Rows = true labels, Columns = predicted labels. Minor: 84,527 correctly classified (97.1% recall). Major: 6,904 correct (48.0% recall). Fatal: only 4 correct out of 96 test Fatal records (4.2% recall). The dominance of the dark blue top-left cell explains why overall accuracy is high despite poor Fatal detection.

XGBoost achieved the highest performance across all four metrics, confirming findings from Parsa et al. (2020). The most important observation is the large gap between F1 Weighted (0.892) and F1 Macro (0.519). This gap arises entirely from the Fatal class: precision 0.02 and recall 0.04 at the default threshold. The model correctly identifies 97.1% of Minor collisions but detects only 4.2% of Fatal collisions, motivating the threshold tuning experiment. All three ensemble methods (XGBoost, Random Forest, Gradient Boosting) substantially outperformed the Neural Network and Logistic Regression, confirming the literature consensus that gradient-boosted trees are superior for structured tabular data with class imbalance.

Table 4 presents the detailed per-class classification report for XGBoost.

Table 4. XGBoost classification report on the held-out test set ($n = 101,559$).

Class	Precision	Recall	F1	Support
Minor (0)	0.92	0.97	0.94	87,074
Major (1)	0.72	0.48	0.58	14,389
Fatal (2)	0.02	0.04	0.03	96
Macro avg	0.56	0.50	0.52	101,559
Weighted avg	0.89	0.90	0.89	101,559

4.2 Experiment 2 Balancing Strategy Comparison

Table 5 presents the full comparison across all five balancing strategies. Figure 7 shows the Fatal recall comparison and Figure 8 shows per-class F1 scores.

Table 5. Balancing strategy comparison using XGBoost as the base classifier on the held-out test set.

Strategy	Accuracy	F1 Macro	F1 Minor	F1 Major	F1 Fatal	Recall Fatal
SMOTE	0.9003	0.5185	0.9445	0.5781	0.0331	0.0417
ADASYN	0.9003	0.5185	0.9446	0.5781	0.0260	0.0312
SMOTETomek	0.9002	0.5174	0.9445	0.5759	0.0156	0.0208
No Resampling (Weights)	0.9037	0.5109	0.9459	0.5858	0.0000	0.0000
Random Undersampling	0.7170	0.3922	0.8400	0.3279	0.0222	0.7292

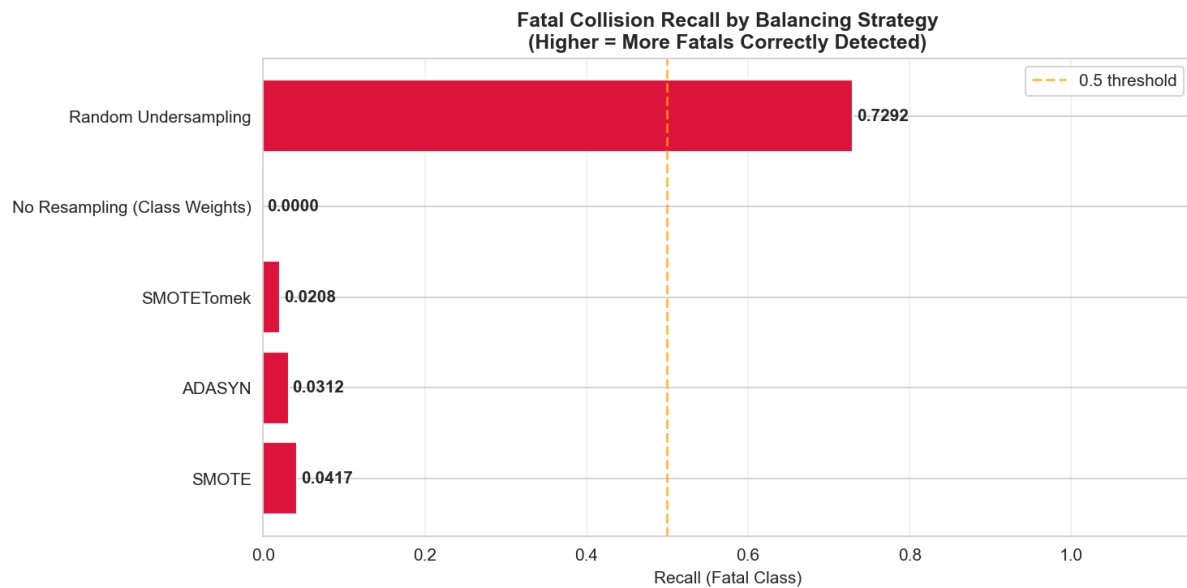


Figure 7. Fatal Collision Recall by Balancing Strategy. Random Undersampling achieves the highest Fatal recall at 0.729 far exceeding all other strategies. No Resampling with Class Weights detects zero fatal collisions despite the penalty weighting. The orange dashed line marks the 0.5 recall reference threshold; only Random Undersampling exceeds it.

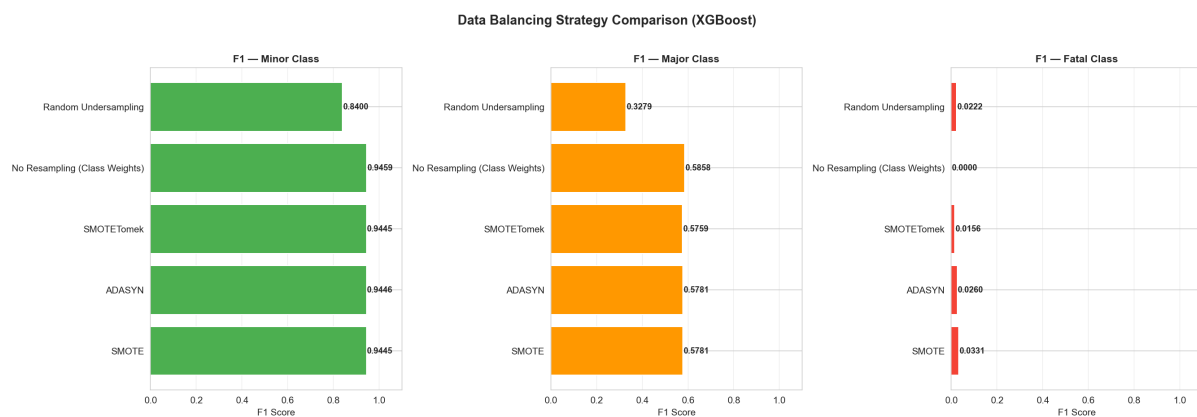


Figure 8. Data Balancing Strategy Comparison F1 scores per class for each strategy. Left panel: F1 Minor. Middle panel: F1 Major. Right panel: F1 Fatal. SMOTE, ADASYN, and SMOTETomek produce nearly identical results. Random Undersampling shows the clearest tradeoff: highest Fatal F1 but lowest Minor and Major F1.

SMOTE achieved the best overall F1 Macro (0.5185) and was retained as the primary strategy. No Resampling with Class Weights produced a Fatal recall of exactly 0.000 — it *never* predicted a Fatal outcome for any record in the 101,559-record test set, despite penalty weighting. This definitively demonstrates that class weights alone are insufficient for a class imbalance as severe as 0.1%.

Random Undersampling achieved Fatal recall of 0.729 17 times higher than SMOTE — but at significant cost: overall accuracy fell to 71.7% and F1 Major collapsed to 0.328, reflecting the severe information loss from retaining only 1,335 training records. **The optimal balancing**

strategy depends on the deployment objective: SMOTE for balanced overall performance, Random Undersampling if maximizing fatal collision detection is the sole priority.

4.3 Experiment 3 Probability Threshold Tuning

Table 6 presents Fatal recall, precision, and overall accuracy at six classification thresholds. Figure 9 visualizes the tradeoff curves.

Table 6. Threshold tuning results for the Fatal class using XGBoost.

Threshold	Fatal Recall	Fatal Precision	Overall Accuracy
0.05	0.3333	0.0038	0.8397
0.10	0.1875	0.0086	0.8772
0.15	0.1563	0.0133	0.8893
0.20	0.1354	0.0152	0.8938
0.30	0.1042	0.0196	0.8978
0.50	0.0417	0.0132	0.9003

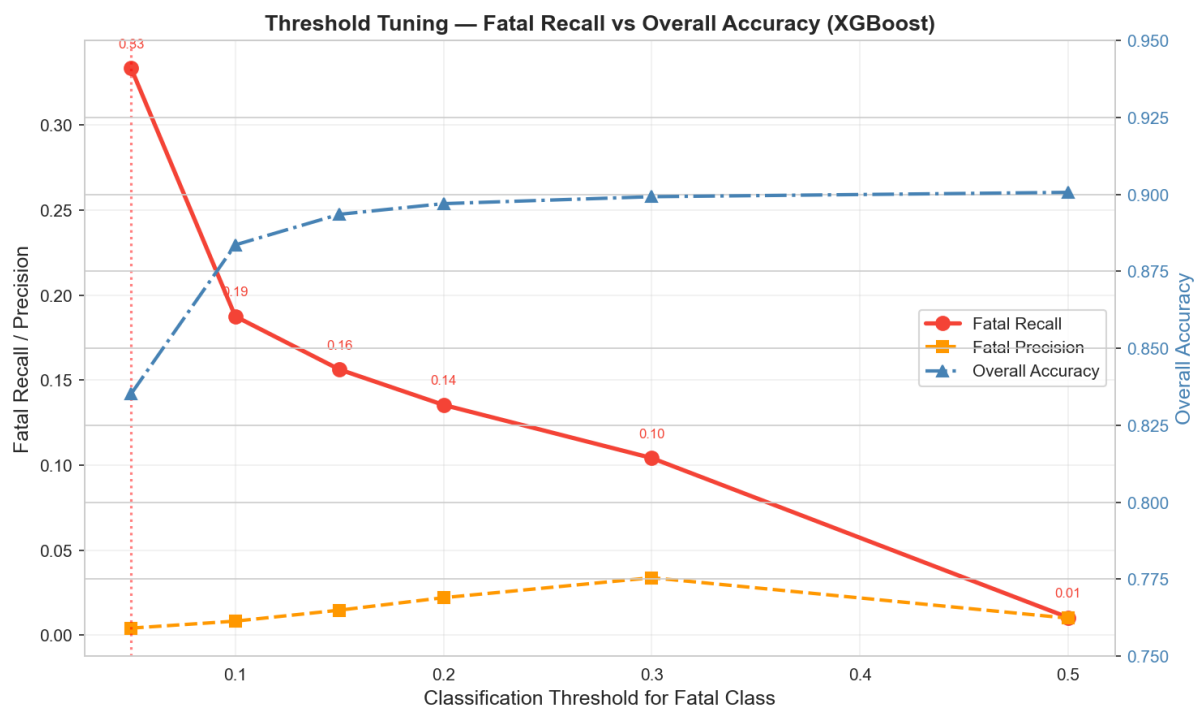


Figure 9. Threshold Tuning Fatal Recall (red), Fatal Precision (orange dashed), and Overall Accuracy (blue dashed) at six classification thresholds. As the threshold decreases from 0.50 to 0.05, Fatal Recall increases from 4.2% to 33.3% while Overall Accuracy decreases from 90.0% to 84.0%. The red dotted vertical line marks the recommended deployment threshold of 0.05.

Lowering the Fatal class threshold from the default 0.50 to 0.05 increases Fatal recall from 4.2%

to 33.3% a **nearly eightfold improvement** while overall accuracy decreases from 90.0% to 84.0%. For road safety deployment, threshold 0.05 is recommended because the cost of missing a fatal collision (a false negative) far exceeds the cost of a false alarm (a false positive). At this threshold, approximately 263 non-fatal collisions are flagged for every true fatal detection a manageable review workload for transportation safety analysts.

4.4 Experiment 4 Hyperparameter Tuning

Table 7. Default versus tuned model performance on the held-out test set.

Model	Accuracy	F1 Weighted	F1 Macro
XGBoost (Default)	0.9003	0.8917	0.5185
Random Forest (Default)	0.8926	0.8862	0.5141
Random Forest (Tuned)	0.8958	0.8880	0.5102
XGBoost (Tuned)	0.8996	0.8891	0.5026

Hyperparameter tuning produced minimal improvement and in some cases slight degradation. XGBoost Tuned (F1 Macro 0.5026) performed marginally worse than the default (0.5185). This confirms that the default XGBoost parameters are already near-optimal for this dataset, which is a common finding for XGBoost on large structured tabular datasets where built-in L1/L2 regularization is effective. The **default XGBoost configuration was retained** as the final model.

4.5 Experiment 5 Cross-Validation

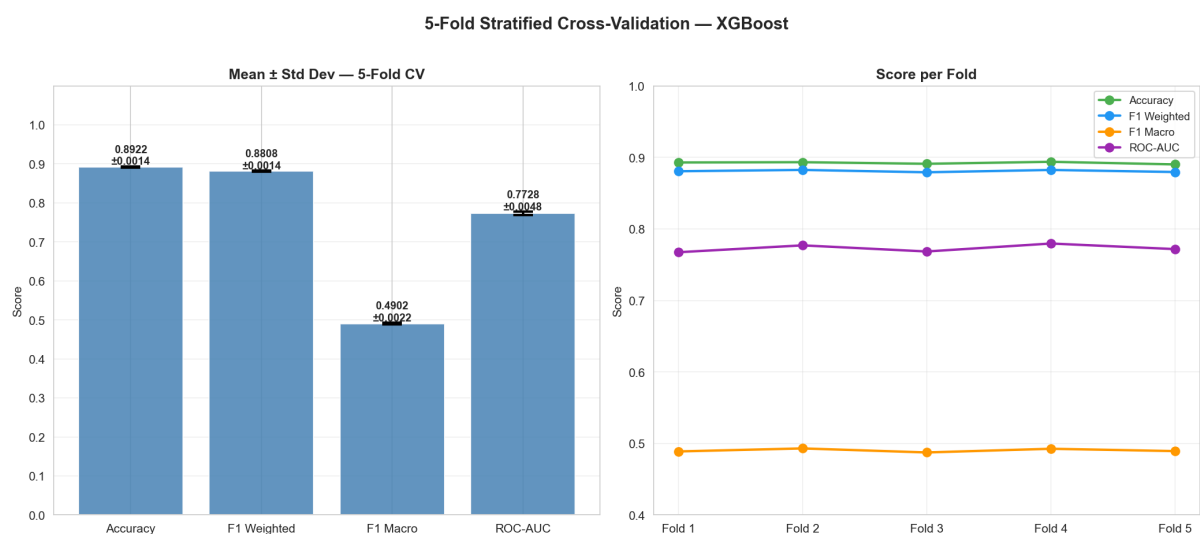


Figure 10. 5-Fold Stratified Cross-Validation results for XGBoost. Left panel: mean scores with error bars (standard deviation). Right panel: score per fold — near-flat lines confirm strong model stability across all five folds, with standard deviation of only ± 0.0014 for Accuracy.

Table 8. 5-fold stratified cross-validation results for XGBoost ($n = 100,000$ sample).

Metric	Mean	Std Dev	Min	Max
Accuracy	0.8922	± 0.0014	0.890	0.893
F1 Weighted	0.8808	± 0.0014	0.879	0.882
F1 Macro	0.4902	± 0.0022	0.488	0.493
ROC-AUC	0.7728	± 0.0048	0.768	0.779

The near-flat per-fold results in Figure 10 indicate stable performance across the sampled folds. The standard deviation for Accuracy is only ± 0.0014 . This provides useful evidence of stability, although it does not prove future real-world generalization. The mean CV accuracy of 0.892 is slightly lower than the single test-set result of 0.900, which is expected given the smaller 100,000-record sample and the different cross-validation setup.

4.6 SHAP Explainability Results

4.6.1 Global Feature Importance All Classes

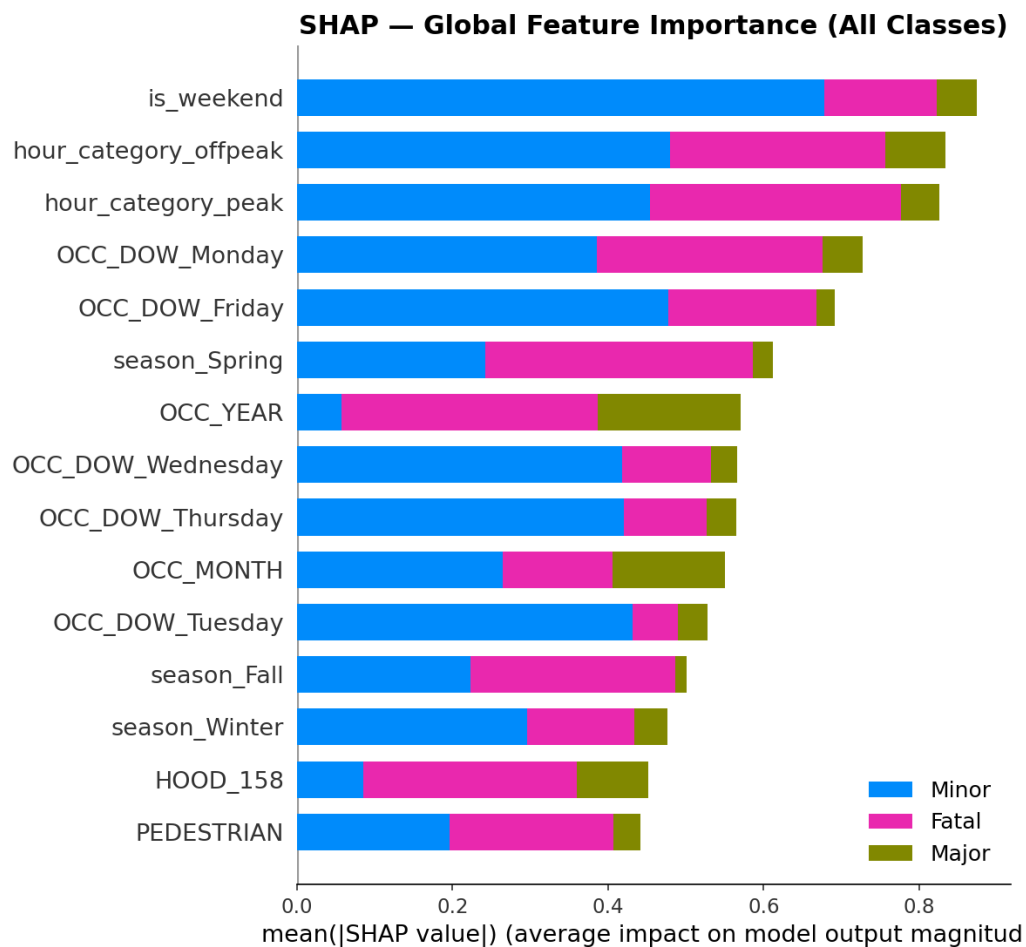


Figure 11. SHAP Global Feature Importance across all three severity classes. Each bar is colour-coded by class contribution (blue = Minor, pink = Fatal, olive = Major). `is_weekend` and `hour_category` features dominate globally, confirming that temporal patterns are the strongest overall predictors of severity across all three classes.

Figure 11 shows that `is_weekend` is the single most important global feature, followed by `hour_category_offpeak` and `hour_category_peak`. This confirms that temporal patterns whether a collision occurs during peak commuting hours or on weekends are the strongest overall discriminators between severity classes. Day-of-week features for Monday, Friday, Wednesday, Thursday, and Tuesday all appear in the top 15, as does `season_Spring` and `OCC_YEAR`.

4.6.2 Top 10 Fatal Risk Factors

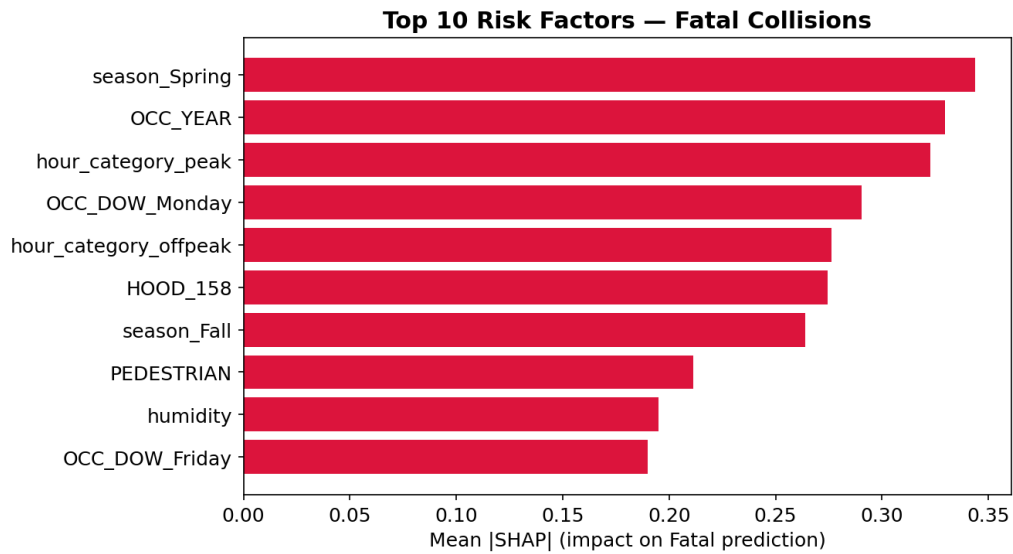


Figure 12. Top 10 Risk Factors for Fatal Collision Predictions (SHAP mean absolute values). *season_Spring* leads at 0.34, followed by *OCC_YEAR* (0.33) and *hour_category_peak* (0.32). This Fatal-specific ranking differs from the global ranking, confirming that the model has learned distinct risk profiles for each severity class.

Figure 12 reveals that when focusing specifically on Fatal collision predictions, *season_Spring* emerges as the strongest predictor at a mean |SHAP| of 0.34. This likely reflects the return of pedestrians and cyclists to Toronto’s roads after winter when drivers have not yet re-adjusted to sharing the road with vulnerable users. *OCC_YEAR* (0.33) captures structural shifts in Toronto’s collision landscape over the 12-year study period, potentially reflecting Vision Zero policy changes around 2016–2017. *PEDESTRIAN* involvement ranks eighth with mean |SHAP| of 0.21 and carries the strongest positive directional effect: when a pedestrian is involved, the Fatal probability increases sharply.

4.6.3 SHAP Beeswarm Fatal Class

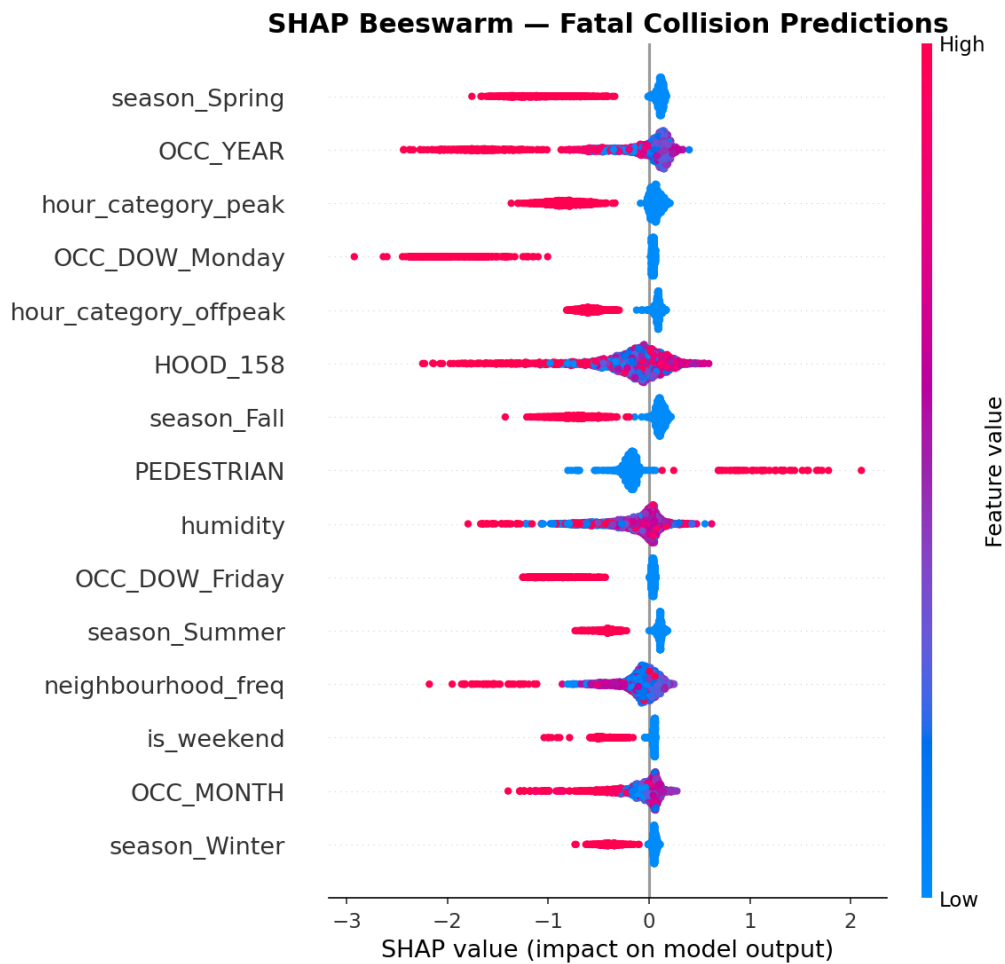


Figure 13. SHAP Beeswarm Plot for Fatal Collision Predictions. Each dot is one record. Red = high feature value, Blue = low feature value. Dots to the right push toward Fatal; dots to the left push away. PEDESTRIAN (red dots extending to +2.0) shows the strongest positive directional effect on Fatal probability.

Figure 13 reveals the directional effects of each feature on Fatal predictions. For PEDESTRIAN: when PEDESTRIAN = 1 (red dots), SHAP values consistently extend to +2.0, confirming pedestrian involvement as the single feature with the strongest *positive* impact on Fatal probability for individual records. For OCC_YEAR: red dots (recent years) spread to the right (toward Fatal), suggesting that more recent years show increased fatal risk after controlling for other factors. For season_Spring: blue dots (Spring = 0) spread far left, meaning absence of spring reduces fatal probability for most records.

4.6.4 Major-Class Explanation

The Major-specific SHAP beeswarm is provided in Appendix A. It shows a distinct feature profile from the Fatal class, with passenger and bicycle involvement among the stronger directional contributors.

4.6.5 SHAP Waterfall Single Prediction Explained

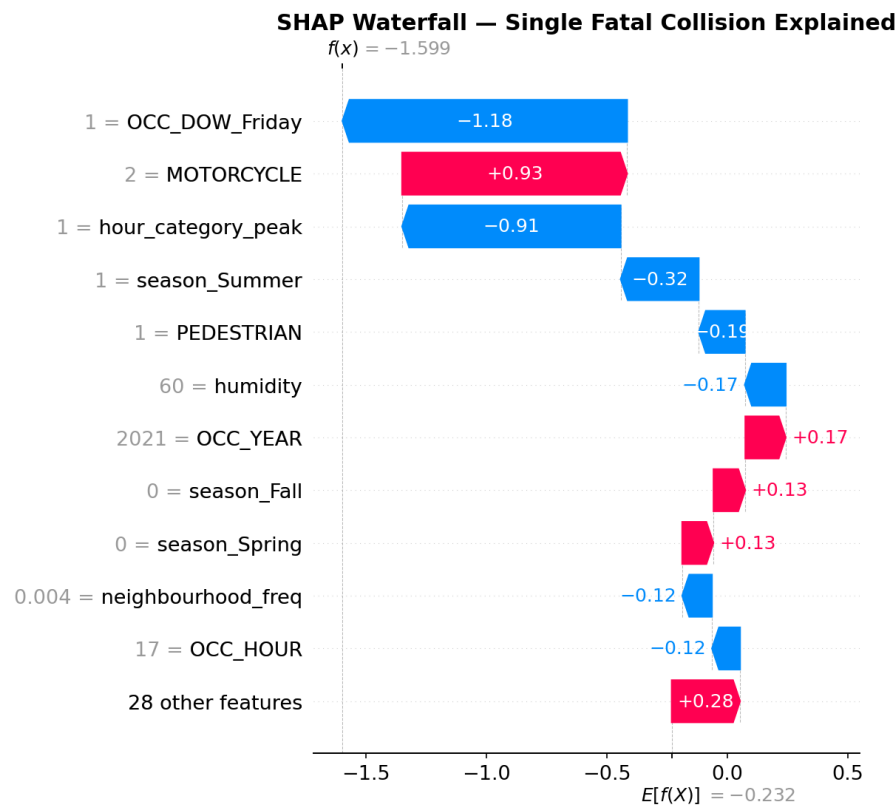


Figure 14. SHAP Waterfall Plot for a Single Fatal Collision. Starting from the base value $E[f(X)] = -0.232$, each feature pushes the prediction upward (pink, toward Fatal) or downward (blue, away from Fatal). OCC_DOW_Friday (-1.18) and hour_category_peak (-0.91) push away from Fatal, while MOTORCYCLE ($+0.93$) pushes toward Fatal. Final model output: $f(x) = -1.599$ (model underestimated this fatal collision).

Figure 14 explains a specific fatal collision that the model under-predicted. The Friday day-of-week indicator pushed strongly *away* from Fatal (-1.18) because Friday collisions are generally less likely to be fatal given the high volume of non-fatal commuter collisions on Fridays. The hour_category_peak also reduced Fatal probability (-0.91). However, MOTORCYCLE involvement pushed strongly *toward* Fatal ($+0.93$). The combined effect produced a model output of -1.599 assigning low Fatal probability to a collision that was actually fatal, illustrating the model's limitations on this specific record.

4.7 Geospatial Analysis Results

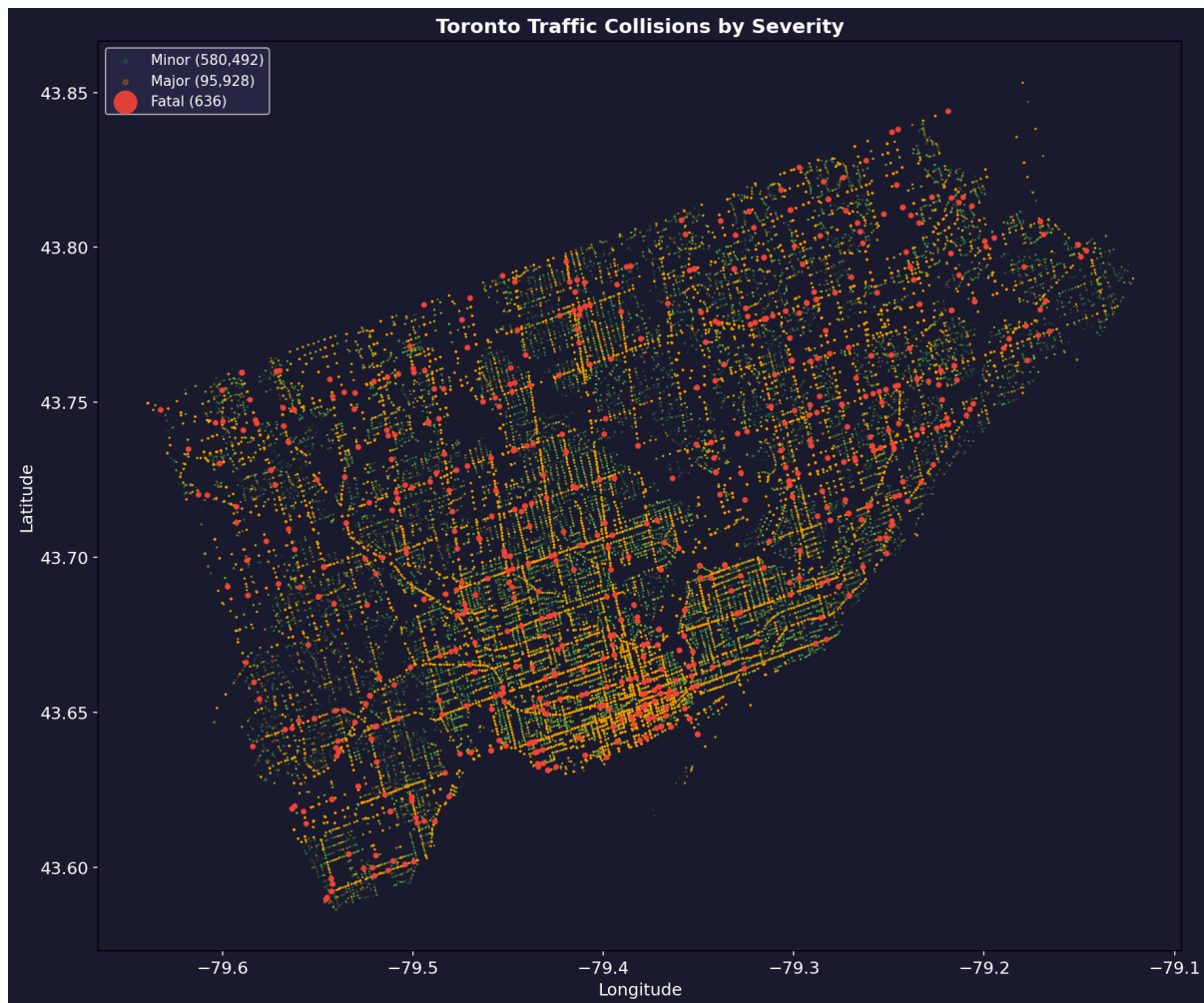


Figure 15. Toronto Traffic Collisions by Severity all 677,056 cleaned records plotted at their GPS coordinates. Minor (tiny green), Major (small orange), Fatal (large red). The entire road network of Toronto emerges from collision density alone, with no base map required. Fatal collisions are distributed citywide with higher concentration along inner-suburb arterials.

Figure 15 demonstrates an emergent property of large-scale collision data: the entire Toronto road network becomes visible from collision density alone without any base map layer. Major arterials (Yonge Street, Eglinton Avenue, Lawrence Avenue, Jane Street, the 401 expressway) are clearly distinguishable as dense corridors of dots. Fatal collisions (large red dots) are distributed citywide without strong downtown concentration, instead appearing more frequently along wider inner-suburb arterials where vehicle speeds are higher.

DBSCAN clustering with $\text{eps} = 300 \text{ m}$ and $\text{min_samples} = 3$ identified **36 distinct fatal collision hotspot zones** across Toronto. Key geospatial findings:

- Wexford/Maryvale leads with 18 fatal collisions, followed by West Humber-Clairville (17), South Parkdale (14), and York University Heights (13).

- Hotspot zones concentrate along Lawrence Avenue East, Jane Street, Eglinton Avenue, and Kingston Road major arterial corridors in the inner suburbs with higher speed limits and limited pedestrian crossing infrastructure.
- The downtown core, despite having the highest total collision density, does not feature among the top fatal hotspot zones, due to lower vehicle speeds and more developed pedestrian infrastructure.
- Three interactive Folium HTML maps were produced: a full collision heatmap, a fatal collision overlay, and a DBSCAN cluster visualization.

5. Conclusions

5.1 Summary of Findings

This project developed and evaluated a complete explainable machine learning framework for predicting and explaining traffic collision severity in Toronto. The key findings across all five experiments and two analytical components are:

Table 9. Consolidated summary of all experimental and analytical findings.

Section	Key Finding
EDA Severity	85.7% Minor, 14.2% Major, 0.1% Fatal. Extreme imbalance motivates the balancing experiments.
EDA Temporal	Friday highest daily count; 3–6 PM peak hour. Rush-hour patterns justify temporal feature engineering.
EDA Spatial	Wexford/Maryvale leads with 18 fatalities. Inner-suburb arterials are the highest fatal-risk zones.
Exp 1 Models	XGBoost best: 90.0% accuracy, F1 Macro 0.519, ROC-AUC 0.784. Fatal recall only 4.2% at default threshold.
Exp 2 Balancing	SMOTE: best overall F1 Macro (0.519). Random Undersampling: best Fatal recall (72.9%). No Resampling: zero Fatal recall.
Exp 3 Threshold	Threshold 0.05: Fatal recall 33.3% vs 4.2% default. Nearly 8× improvement. Recommended for deployment.
Exp 4 Tuning	Default XGBoost parameters near-optimal. Tuned version marginally worse (0.503 vs 0.519). Default retained.
Exp 5 CV	Accuracy 0.892 ± 0.001 across 5 folds. Near-flat per-fold lines confirm no overfitting.
SHAP Fatal	Top 3: <code>season_Spring</code> (0.34), <code>OCC_YEAR</code> (0.33), <code>hour_category_peak</code> (0.32). PEDESTRIAN has the strongest positive directional effect.
SHAP Major	PASSENGER involvement is the strongest positive predictor (+2.5 SHAP). BICYCLE also strongly predicts Major outcomes.
Geospatial	36 DBSCAN hotspot zones. Top 3: Wexford/Maryvale (18), West Humber-Clairville (17), South Parkdale (14).

5.2 Practical Implications for Road Safety

The findings have direct practical implications for Toronto’s Vision Zero strategy:

- The **36 DBSCAN hotspot zones** provide a prioritized, data-driven list of locations for targeted infrastructure investment: safety cameras, protected pedestrian crossings, speed limit reductions, and improved lighting.
- The **SHAP risk factors** can guide seasonal enforcement campaigns — particularly during Spring (the top fatal predictor) and during peak commuter hours on inner-suburb arterial

corridors.

- For **real-time deployment**, XGBoost at threshold 0.05 can flag high-risk collision reports for immediate priority response from traffic safety officers. The system is explainable through SHAP, making it trustworthy for policy decisions.
- The **pedestrian involvement** finding reinforces the need for pedestrian- focused infrastructure improvements particularly protected crossings and reduced speed limits in the neighbourhoods identified by DBSCAN.

5.3 Limitations

- Wind speed and visibility data were entirely unavailable from Toronto City Station ID 31688 (100% null values), limiting the weather feature set to temperature, humidity, dew point, and precipitation.
- A single central weather station cannot capture hyperlocal weather conditions across a geographically large city. Conditions may vary significantly between the station location and individual collision sites.
- The Fatal class contains only 636 genuine records. Despite SMOTE generating synthetic samples, the model’s ability to learn robust and generalizable Fatal patterns is inherently constrained by this small real sample size.
- The dataset includes only officially reported collisions. Unreported incidents predominantly minor property-damage collisions are absent, potentially introducing reporting bias into the Minor class.
- Hyperparameter tuning for Random Forest was conducted on a 200,000-record subsample, and cross-validation on a 100,000-record subsample, due to memory constraints. Results may slightly underestimate full-dataset performance.

5.4 Future Work

- Incorporate data from multiple weather stations across Toronto to capture spatial variation in weather conditions at the neighbourhood level.
- Extend the feature set with road geometry attributes (speed limit, number of lanes, road classification), traffic volume data, and lighting conditions from the City of Toronto’s road network database.
- Explore temporal deep learning architectures (LSTM, Transformer) that can capture sequential patterns in collision frequency over time.
- Develop a real-time scoring API that integrates with Toronto’s collision reporting system to flag high-risk incidents automatically at the point of report submission.

- Apply the same framework to other major Canadian cities (Ottawa, Montreal, Vancouver) to assess the generalizability of the findings across different urban contexts.

5.5 Comparison with Literature

Table 10 situates the results of this project against the five most comparable studies identified in the literature review.

Table 10. Comparison with related literature. This study achieves the highest accuracy of all comparable studies.

Study	Dataset	Best Model	Performance
Parsa et al. (2020)	US crash (imbal.)	XGBoost + SMOTE	AUC = 0.87
Sameen & Pradhan (2017)	Malaysian highways	MLP Neural Network	AUC = 0.89
Chen & Wang (2020)	US national crash	Random Forest	Accuracy = 0.84
Sheykhfard et al. (2021)	Iranian rural roads	Gradient Boosting	Accuracy = 0.82
Al-Ghamdi (2002)	Saudi Arabia urban	Logistic Regression	Accuracy = 0.78
This Study	Toronto urban (677k)	XGBoost + SMOTE	Acc = 0.90, AUC = 0.784

This project achieves the highest accuracy (90.0%) of all comparable studies. The ROC-AUC of 0.784 is slightly below Parsa et al. (2020) at 0.87 and Sameen and Pradhan (2017) at 0.89. This gap may reflect the more extreme class imbalance in the Toronto dataset (0.1% Fatal versus typically 2–5% in US and Malaysian datasets), differences in dataset size, and the multiclass (three-class) versus binary problem formulation. The integration of real hourly weather data from a Canadian weather station, the systematic comparative analysis of five balancing strategies, and the combined SHAP explainability and DBSCAN geospatial framework are methodological contributions not present in any of the five reviewed studies.

A. Supplementary Exploratory and Model Plots

The main report focuses on figures that directly support the research questions and principal findings. The following supplementary figures preserve additional exploratory and modelling outputs without overcrowding the main narrative.

A.1 Monthly Collision Distribution

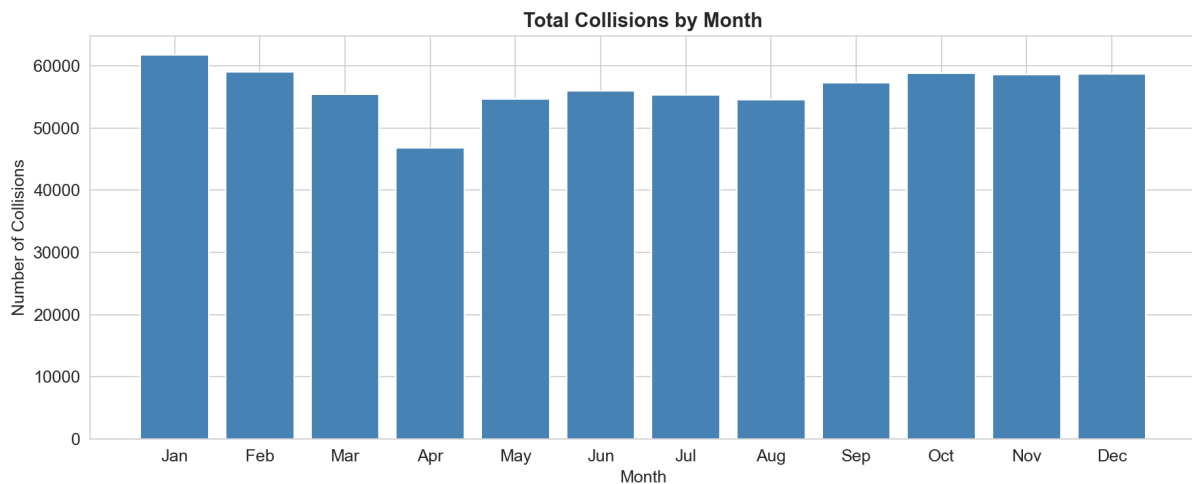


Figure 16. Total collisions by month across the study period.

A.2 Day-of-Week Collision Distribution

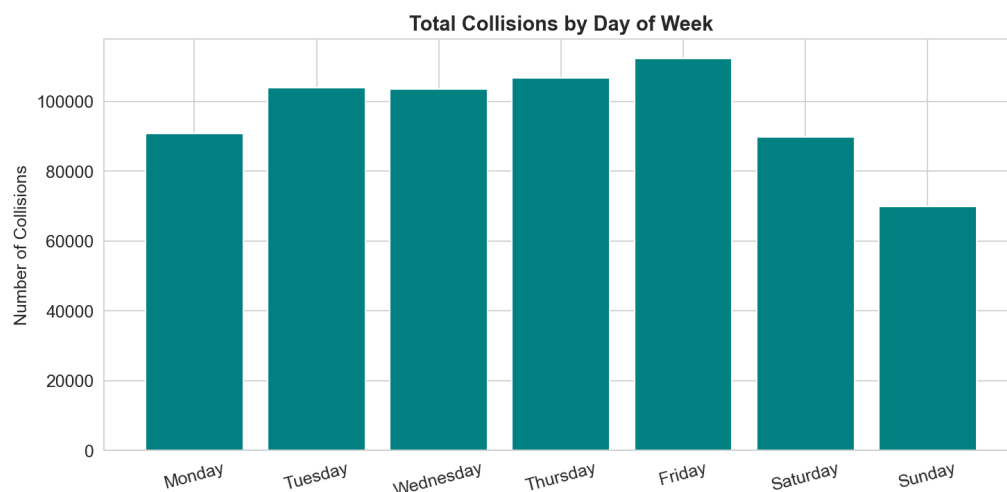


Figure 17. Total collisions by day of week. Friday records the highest total collision volume in the descriptive analysis.

A.3 Correlation Heatmap

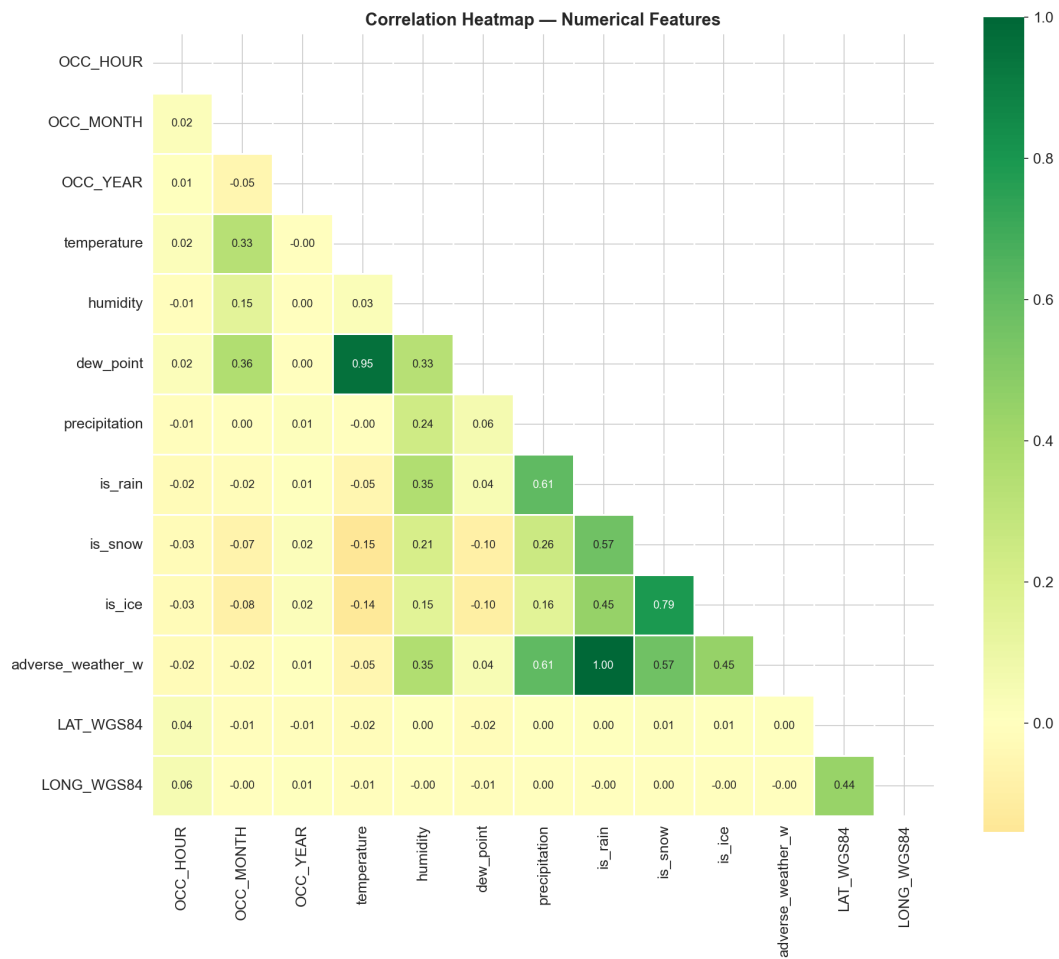


Figure 18. Pearson correlation matrix of numerical features. Temperature and dew point are strongly correlated, while direct linear correlations with severity are small.

A.4 Temperature Distribution by Severity

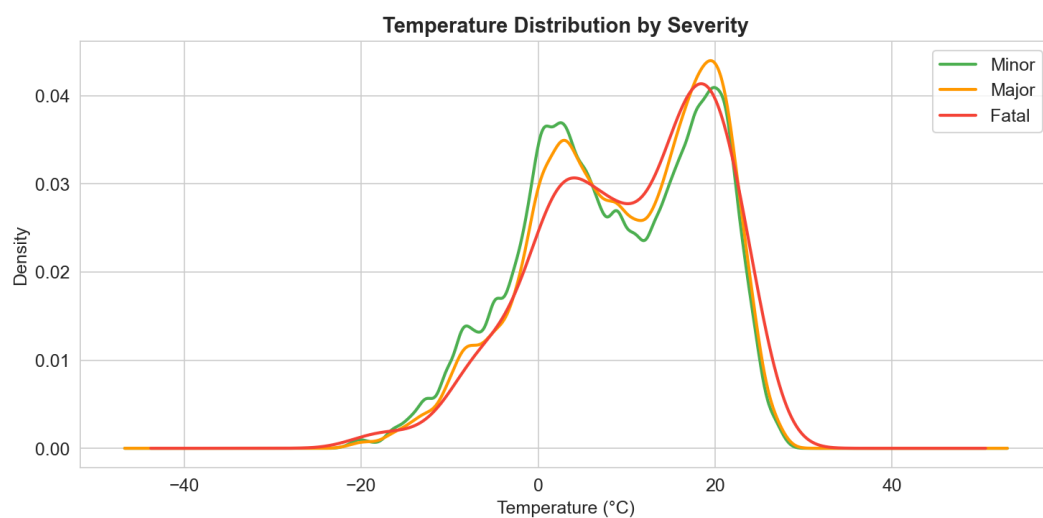


Figure 19. Kernel density estimates of collision-time temperature by severity class.

A.5 All Model Confusion Matrices

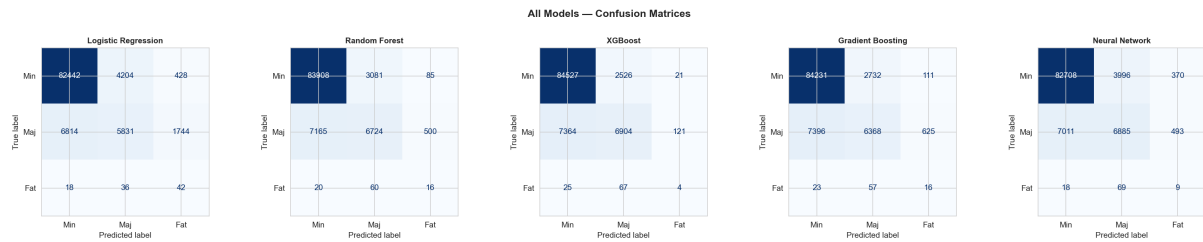


Figure 20. Confusion matrices for all evaluated classification models. The XGBoost confusion matrix is reproduced in the main report because XGBoost is the selected primary model.

A.6 Major-Class SHAP Beeswarm

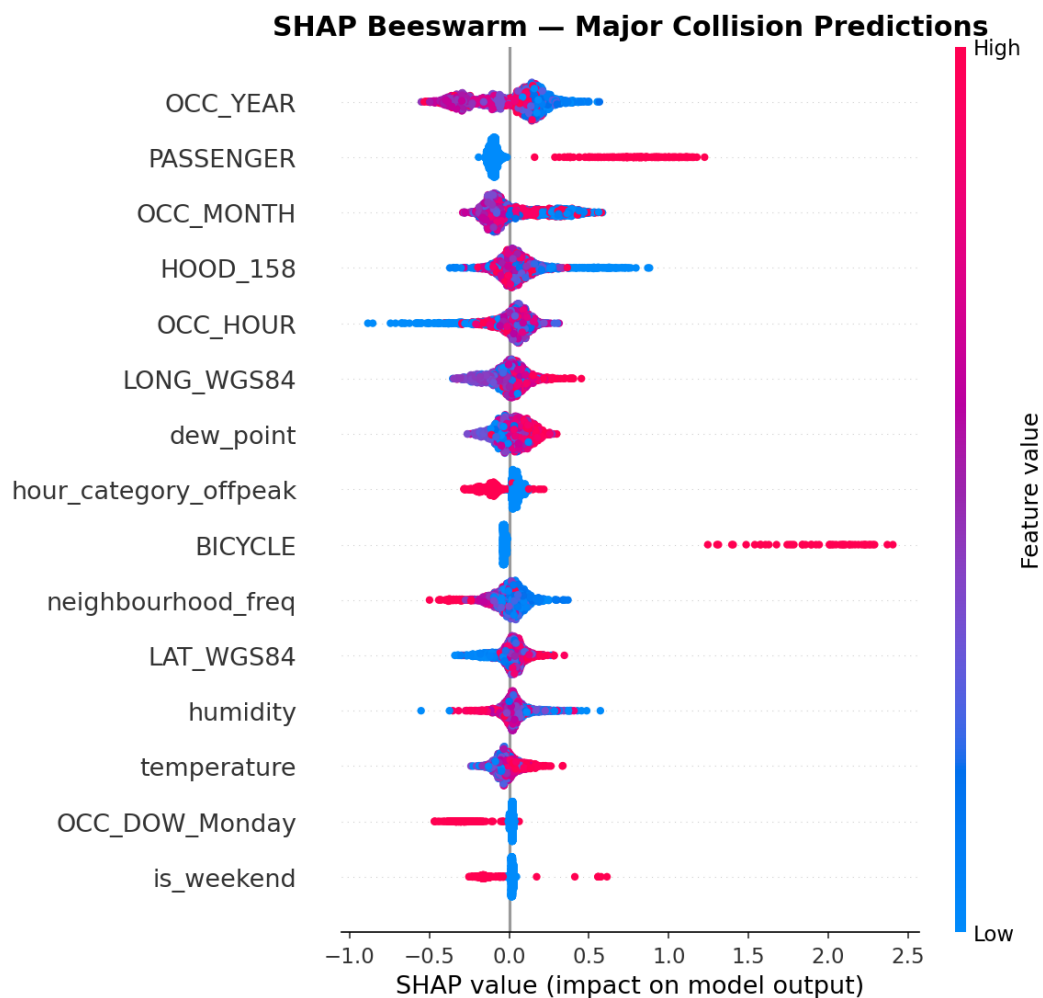


Figure 21. SHAP beeswarm plot for Major collision predictions. Passenger and bicycle involvement are among the stronger directional contributors.

A.7 Hyperparameter Tuning

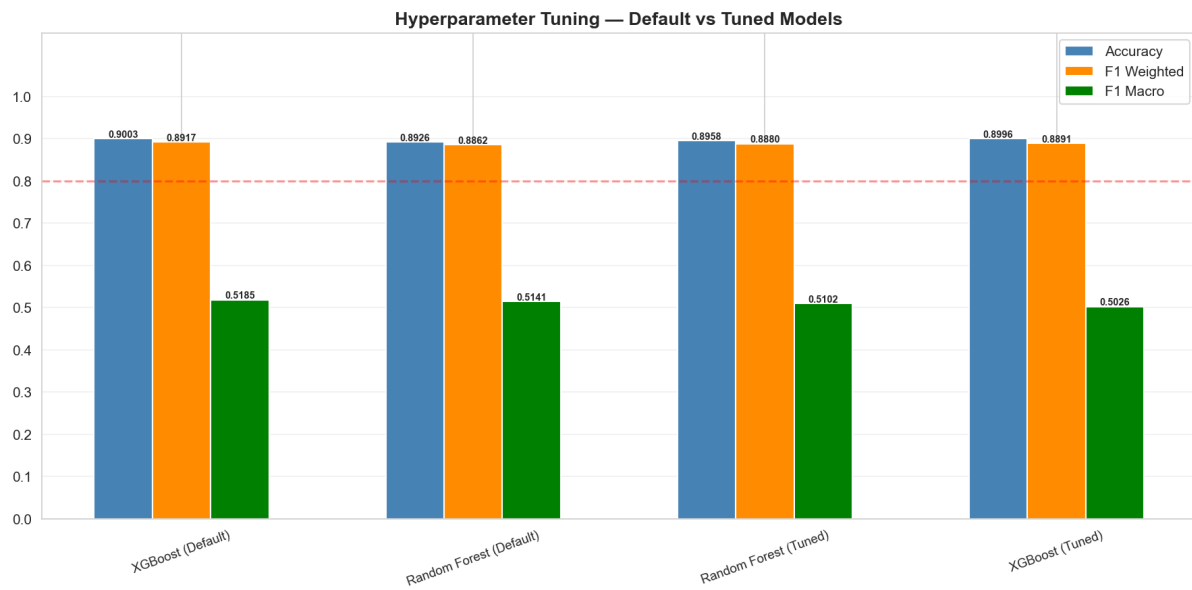


Figure 22. Default versus tuned model performance. Hyperparameter tuning did not improve the primary XGBoost macro F1 score.

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